25 A force of 70 newtons acts on an area of 20 cm^2

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The force is increased by 10 newtons.
The area is increased by 10 cm^2

Helen says,

"The pressure decreases by less than 20%"

Is Helen correct?
You must show how you get your answer.

A = area in cm^2; F = force in N.

	A	F	
	20	70	
÷20			÷20
	1	3.5	

Old pressure = 70 ÷ 20 = 3.5 N/cm^2

1 mark
Find a pressure using force ÷ area

New force = 80 N; new area = 30 cm^2.

$$\text{New pressure} = \frac{80}{30} = \frac{8}{3} \approx 2.6667 \text{ N/cm}^2$$

Pct = percentage; p = old pressure in N/cm^2.

	Pct	p	
	100	3.5	
÷10			÷10
	10	0.35	
×2			×2
	20	0.7	

20% of 3.5 = 0.7

After 20% decrease: 3.5 – 0.7 = 2.8

1 mark
Compare with a 20% decrease

2.6667 < 2.8, so the decrease is more than 20%.

No, Helen is not correct.

1 mark
Correct, supported decision

(Total for Question 25 is 3 marks)